



Response to Request for Information: AI Action Plan Development

I. Introduction

Artificial Intelligence (AI) is at an inflection point in government and national security applications. While agencies recognize AI's transformative potential, traditional approaches remain constrained by manual data labeling, which leads to slow, costly, and unscalable AI deployments. AI models struggle to achieve the precision, adaptability, and governance required for mission success without a data-first approach. To bridge this gap, government agencies must embrace programmatic data development—an approach that accelerates AI model training, ensures transparency, and enables real-time adaptation to evolving threats and operational needs.

About Snorkel AI

Snorkel AI has pioneered programmatic data labeling, leveraging weak supervision to create high-quality AI training data at scale. Unlike traditional manual approaches, Snorkel Flow enables Subject Matter Experts (SMEs) and data scientists to collaborate programmatically, slashing the time and cost of AI development while ensuring models are mission-ready, adaptive, and compliant. By shifting the focus from model-centric AI to scalable, data-centric strategies, Snorkel AI empowers organizations to rapidly deploy AI solutions that are accurate, transparent, and continuously improving.

Founded out of the Stanford AI Lab, Snorkel AI is revolutionizing AI data development with its patented weak supervision technology. This technology programmatically integrates multiple noisy data sources to generate precise, labeled datasets. This software-driven approach replaces slow, manual annotation, allowing government agencies and enterprises to accelerate AI adoption by 10 to 100x while reducing risk and cost. Already deployed across Fortune 500 companies and national security agencies, Snorkel AI ensures AI systems remain agile, auditable, and optimized for real-world impact.

II. Executive Summary

Regardless of their sophistication, AI models are fundamentally constrained by the quality, completeness, and representativeness of the data they are trained on. Yet, current AI development efforts often treat data as an afterthought—resulting in slow, expensive, and unreliable deployments that fail to scale. National security agencies and government organizations cannot afford to build AI systems on incomplete, biased, or static datasets. In high-stakes environments, poor data quality directly translates to mission failure.

The U.S. Government (USG) must adopt a data-first AI strategy that prioritizes scalable, transparent, and adaptive data pipelines rather than introducing new risks or inefficiencies to ensure AI enhances decision-making and operational agility. This means moving beyond traditional, manual data-labeling approaches and embracing programmatic, scalable methods that enable AI systems to continuously learn, adapt, and improve. This document outlines a practical, mission-driven AI action plan that prioritizes governance, adaptability, and operational readiness—ensuring that AI initiatives deliver reliable, transparent, and high-impact results at scale.

This AI action plan provides four key recommendations to ensure AI scalability, governance, and operational effectiveness:

1. Shift from Model-Centric to Data-Centric AI – AI deployment must be accelerated through scalable data pipelines that reduce reliance on slow, manual data-labeling efforts.
2. Strengthen AI Governance and Compliance – A robust AI data governance framework ensures auditability, security, and bias mitigation across all AI applications.
3. Implement Reinforcement Learning & Agentic Workflows – AI systems must continuously learn and adapt, requiring a data-first policy framework that integrates real-time feedback loops and mission-driven decision-making.
4. Adopt Fine-Grained, Programmatic AI Evaluation – AI models should be evaluated continuously using automated techniques that enable mission-specific performance assessments, real-time adaptation, and rigorous transparency mechanisms.

By adopting this data-first AI strategy, the USG can accelerate AI deployment, enhance decision-making, reduce costs, and ensure AI applications deliver mission success at scale.

III. The Problem: AI's Effectiveness is Limited by Data, Not Model Sophistication

For years, AI development has been dominated by a model-first approach, where organizations focus on fine-tuning architectures while treating data as an afterthought. This approach has repeatedly failed to deliver reliable AI outcomes in commercial and government environments. The reality is that AI models are only as effective as the data they are trained on. Yet, many organizations struggle with accuracy, scalability, and governance in AI development. Even the most advanced models cannot overcome deficiencies in training data. Poor-quality, biased, or incomplete datasets introduce systemic vulnerabilities, increasing the likelihood of model hallucinations, incorrect predictions, and decision failures in mission-critical environments.¹

One of the most significant risks AI faces today is accuracy and adaptability. Models trained on incomplete or low-quality data are prone to hallucinations, misinformation, and biased decision-making, which is especially dangerous in high-stakes environments like national security, intelligence analysis, and cyber defense.² Additionally, model drift—where AI systems degrade over time due to changing mission needs or unseen data distributions—further diminishes reliability. The inability of traditional AI pipelines to rapidly adapt to evolving threats and mission requirements means that AI models often become obsolete before they can provide meaningful operational value.

The inability to scale AI training and evaluation processes remains a major bottleneck. Traditional AI data curation methods—such as manual labeling, outsourced annotation, or crowdsourcing—introduce significant bottlenecks, making it difficult to evaluate, fine-tune, and scale AI models efficiently.³ Many AI projects fail to move beyond the proof-of-concept stage because the time and cost required to curate, label, and structure training data for production-grade performance are too high. This is particularly evident in enterprise and government applications, where AI systems must process massive volumes of complex,

¹ Soni, A., Arora, C., Kaushik, R., & Upadhyay, V. (2023). Evaluating the Impact of Data Quality on Machine Learning Model Performance. *Journal of Nonlinear Analysis and Optimization*, 14(1), 8-18.

² Hu, M., Behar, E., & Ottenheimer, D. (2024). National Security and Federalizing Data Privacy Infrastructure for AI Governance. *William & Mary Law School Scholarship Repository*, 1829.

³ Renggli, C., Rimanic, L., Gürel, N. M., Karlaš, B., Wu, W., & Zhang, C. (2021). A Data Quality-Driven View of MLOps. *arXiv preprint arXiv:2102.07750*.

domain-specific data. Organizations cannot operationalize AI effectively without a systematic, scalable approach to data development.

Governance and compliance challenges further obstruct AI adoption, particularly due to the lack of version-controlled datasets, explainability frameworks, and structured audit trails. The inability to audit AI training data increases regulatory risk, eroding trust in AI-driven decision-making.⁴ Security vulnerabilities arise when AI models are trained on unverifiable data, opening the door to adversarial manipulation, bias, and misinformation. Additionally, the lack of version control in AI training data prevents organizations from tracking changes, ensuring consistency, and mitigating operational risks. This lack of traceability in regulated industries and government applications makes meeting transparency and accountability requirements nearly impossible.

The common denominator across all these challenges is that AI is not just about building better models but also better data. The key to unlocking AI's full potential lies in programmatic data development, which elevates AI training from a manual, error-prone process to a scalable, systematic approach. This shift allows organizations to generate high-quality, structured, and auditable training data at a fraction of the time and cost required by traditional methods. At Snorkel AI, we have spent the last decade pioneering this approach, enabling AI teams to move beyond static, manually labeled datasets and instead develop, refine, and adapt their data programmatically.

As organizations navigate the next phase of AI deployment, those relying on outdated, model-first approaches will struggle to transition from flashy demos to production-ready AI. The following sections will outline the solutions that enable AI to be trained, evaluated, and deployed at mission scale—without reliance on inefficient manual processes. In an era where data is the new frontier in AI, enterprises and government agencies must recognize that their success or failure in AI will ultimately depend on how they develop their data.

IV. The Solution: A Data-First AI Strategy for the U.S. Government

⁴ National Security Council. (2024). Framework to Advance AI Governance and Risk Management in National Security.

1. Shift Focus from Model-Centric AI to Data-Centric AI: Accelerating Specialized AI Development by Scaling Domain Expertise

The most significant strategic advantage of AI is access to high-quality, mission-aligned data. In a world where advanced AI models have been commoditized, the true differentiator is the ability to specialize AI models using proprietary data to drive mission success.

From Model-First to Data-First AI Development

- “Generalist” AI models are becoming increasingly commoditized. Specialization is required for mission-critical applications, which require access to proprietary data for training and evaluation.
- Agencies must transform their data to be structured, labeled, and task-oriented. Data quality and readiness should be prioritized over model selection.
- Auditability & Traceability: AI decisions must be fully traceable, with provenance metadata, lineage tracking, and explainability mechanisms embedded into programmatic data pipelines.
- To achieve the data quality required for specialization, AI initiatives should mandate that experts are kept in the loop. Mission success depends on programmatically applying subject matter expertise and avoiding manual labeling by non-experts

Eliminating Bottlenecks in AI Data Development

The primary roadblock to AI deployment is the reliance on slow, manual data labeling processes, which are costly, time-consuming, and unscalable.

- Current approach: Manual annotation—slow, high-cost, and unscalable.
- Proposed solution: Programmatic data labeling, enabling rapid iteration, continuous learning, and scalable AI fine-tuning.⁵

To accelerate the development of specialized models, we recommend the Administration:

- Ensure that the federal government has access to software capabilities that allow subject matter experts to programmatically transform unstructured data into high-quality datasets suitable to power the next generation of specialized AI

⁵ Ratner, A., Ehrenberg, H., Hussain, Z., Dunnmon, J., & Ré, C. (2019). *Weak Supervision: The New Paradigm for Machine Learning*. Stanford AI Lab.

- Direct federal agencies to curate and develop their proprietary data programmatically, with the ambition to develop highly specialized task-oriented AI systems

2. Strengthen AI Governance, Compliance, and Transparency

To ensure AI is trustworthy and mission-aligned, the USG must implement a robust AI Data Governance Framework with four key pillars:

- **Auditability & Traceability:** AI decisions must be fully traceable to specific labeled data and programmatic data pipelines.
- **Bias & Security Controls:** AI models must be trained on balanced, mission-relevant datasets to mitigate bias, adversarial risks, and unauthorized access.
- **Adaptability:** AI systems should ingest new mission data in real-time without requiring full retraining.
- **Secure AI Infrastructure:** AI systems must adhere to zero-trust principles, ensuring role-based access control (RBAC), cryptographic data integrity checks, and federated learning architectures for decentralized data security.

By ensuring transparency, security, and adaptability, the USG can deploy reliable, mission-ready AI while reducing risk.

3. Reinforcement Learning & Agentic Workflows: A Data-First Policy Framework

Reinforcement learning (RL) and data-driven AI methodologies complement one another in a variety of significant ways:

Data Quality as the Foundation for RL Success

Reinforcement learning agents learn by interacting with their environment and feedback loops, but such incremental learning is exceedingly sensitive to initial data quality:

- Ground-truth baseline data gives the starting state representations and reward structures that inform agent learning
- Calibration data sets suitable baselines for evaluating agent performance
- Good-quality historical data supports good pre-training prior to live deployment

Without robustly curated foundation data, RL agents are vulnerable to reward misspecification, overfitting to training distributions, and adversarial exploitation - learning to

optimize the wrong goals or discovering new shortcuts that technically meet rewards but carry no genuine mission intent.⁶

Complementary Feedback Loops

Data-driven RL and AI generate strong complementary feedback loops:

- Improved RL with data curation: Well-structured, clean data enables RL agents to learn more about the state spaces and consequences of actions
- Improved collection of data with RL: Well-planned agents can recognize gaps in data and look for useful data
- Continuous Learning: AI agents must be designed to dynamically incorporate new mission-relevant data, with mechanisms to detect and mitigate drift in learned policies.

Overcoming Compounding Error Challenges

Reinforcement learning systems have some specific error propagation challenges:

- Error compounding: Minor initial biases in data can cause disastrous compounding errors as agents keep adding upon their learned policies
- Drift detection: Agents must have strong monitoring to identify when agent behavior has moved away from anticipated patterns
- Intervention mechanisms: Human monitoring and backup plans must be activated when data quality declines

Designing Resilient Agentic Workflows

In government environments, agentic workflows must be designed with mission resiliency in consideration:

- Data provenance tracing: All data sources have to be logged and checked for quality
- Multi-modal verification: Key decisions have to be checked against different modes and data sources
- Domain-specific guardrails: Mission parameters have to limit agent exposure to avoid unsafe action
- Graceful degradation: Systems have to deal with data quality problems by autoionizing less instead of carrying on with bad inputs

USG Implementation Suggestions

To implement these systems effectively, the USG has to take into account the following:

- AI development teams paired with committed data engineering teams

⁶ Budach, L., Feuerpfeil, M., Ihde, N., Nathansen, A., Noack, N., Patzlaff, H., Harmouch, H., & Naumann, F. (2022). The Effects of Data Quality on ML-Model Performance. Proceedings of the VLDB Endowment, 14(1).

- Cross-agency data standards to enable knowledge sharing across different mission areas
- Rich simulation environments modeled from great data for agent testing in a safe and controlled environment before deployment
- Gradated autonomy architectures that enable increasingly autonomous agents as performance criteria are achieved
- Routine adversarial testing to identify potential failure modes before deployment

The Path Forward

The eventual effectiveness of AI systems like the future government relies on holding data infrastructure as mission-critical. Merging reinforcement learning with data-driven and agentic practices allows agencies to create systems that:

- Improve incrementally without compromising alignment with mission goals
- Learn to adapt to shifting conditions without threatening operational dependability
- Offer transparent explanation for action required by data lineage
- Guard against adversarial manipulation through strong validation procedures

This strategy reorganizes the traditional image of data as a static commodity into data as a dynamic source for more autonomous systems.

4. Implement Fine-Grained, Programmatic Evaluation of AI Models

To ensure AI reliability, adaptability, and mission alignment, the USG must adopt a comprehensive AI evaluation framework that moves beyond static performance benchmarks and enables continuous, mission-driven assessment.

Traditional AI evaluation relies on static test sets and one-time SME reviews, which lack the granularity, automation, and mission-adaptive feedback loops required for real-world deployments. Instead, the USG should implement a systematic, data-centric evaluation framework that:

- Continuously assesses model performance across operational scenarios.
- Use programmatic techniques (e.g., weak supervision, LLM-as-a-judge, fine-grained slicing) to identify and correct AI failure modes rapidly.
- Provides metrics on how well the above programmatic techniques align with SME knowledge.

- Establishes continuous, mission-specific AI evaluation pipelines, integrating real-time adversarial testing and performance degradation monitoring to ensure sustained model integrity.
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V. Conclusion: Data as the Foundation of AI Success

The future of AI in government and national security depends on how well the USG develops, structures, and governs its data. While past AI initiatives have focused on improving model architectures, real operational impact will come from scalable, transparent, continuously evolving data pipelines.

To achieve this, the USG must take immediate action:

1. Adopt a Data-First AI Development Framework – AI models can only be effective if trained on high-quality, continuously curated, and programmatically adaptable data aligned with evolving mission priorities.
2. Accelerate AI Deployment Through Scalable Data Pipelines – AI must move beyond pilot projects. Programmatic data labeling and iterative refinement will enable faster, more effective AI deployments across government agencies.
3. Embed SME Knowledge into AI Systems – AI systems must integrate subject matter expertise into their training data to ensure models remain mission-aligned and adaptable.
4. Strengthen AI Governance and Evaluation—AI reliability depends on fine-grained, programmatic evaluation that ensures auditability, bias mitigation, and real-time adaptability to evolving mission needs.

A data-centric AI approach is not just a technological shift but an operational necessity. By prioritizing scalable data infrastructure, automated governance, and mission-adaptive evaluation, the USG can rapidly transition AI from experimental to operational at scale.

At Snorkel AI, we believe that the future of AI in national security and government hinges on a data-first approach, one that prioritizes scalability, adaptability, and governance from the ground up. Our work with leading enterprises and government agencies has proven that programmatic data development is the key to unlocking AI's full potential while mitigating

risk, reducing costs, and accelerating deployment. As AI continues to reshape mission-critical operations, ensuring policies reflect the importance of scalable, high-quality data will be paramount to maintaining our competitive and security advantage.

I would welcome the opportunity to serve as a consultative resource in shaping this critical policy and ensuring the U.S. Government has the tools and strategies necessary to lead in AI. If there is a way I can support your efforts—whether through sharing insights, facilitating discussions, or providing technical guidance—I am at your disposal. I’m excited to work together to define a data-first AI strategy that secures the future of our nation’s AI capabilities.

Signed,

Alex Ratner
Founder & CEO, Snorkel AI
Affiliate Assistant Professor, University of Washington
[REDACTED]
55 Perry St
Redwood City, CA 94063